Simple and multiple regression models estimation

Econometrics. Lecture 2

Content

1. Simple regression model
2. Multiple regression model
3. Assumptions of the classical linear regression model
4. Units of measurement and functional form

1. Simple regression model

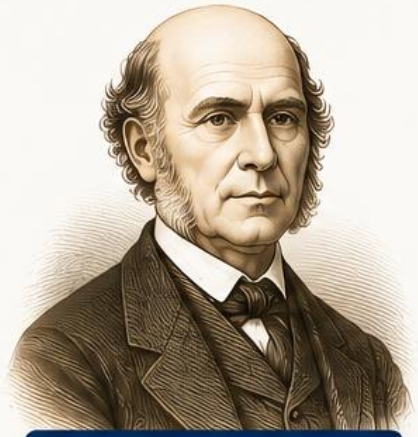
Regressions: an economist's obsession

- Countries that trade more with the rest of the world also have higher income levels – but does this mean that trade raises income levels?
- People with more education tend to have higher earnings, but does this imply that education results in higher earnings?

Regression is a basic statistical tool for distinguishing between correlation and causality.

THE ORIGINS OF REGRESSION

From Francis Galton's observations to modern statistics



SIR FRANCIS GALTON (1822–1911)

Cousin of Charles Darwin, pioneer in the study of heredity, evolution and statistics.

WHY ANIMALS AND PLANTS?

Galton was fascinated by how traits are inherited.
He collected and analyzed data on:



- **Plants:** sweet pea seeds (size, shape, etc.)



- **Dogs:** sizes and other characteristics



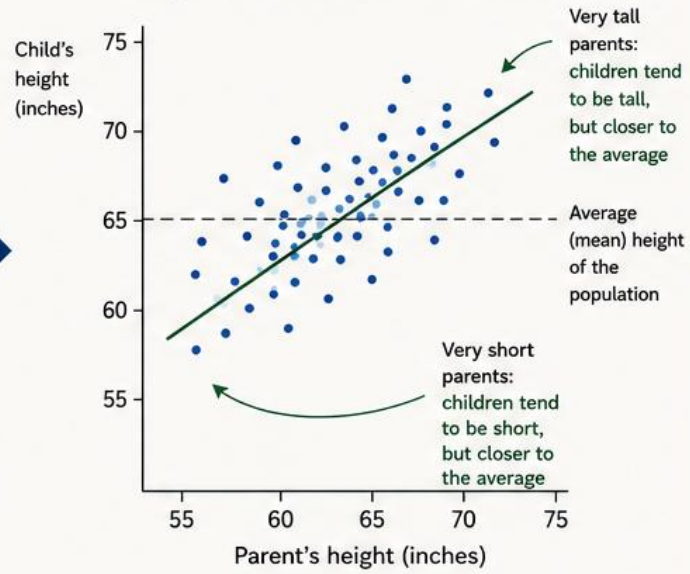
- **Humans:** height, skull measurements, and more

WHAT DID GALTON OBSERVE?

When he studied human height and pea seed size, Galton noticed a remarkable pattern:

Extreme values in parents tend to be closer to the average in their children.

Example: Height of Parents vs. Children



ALSO OBSERVED IN PEA SEEDS

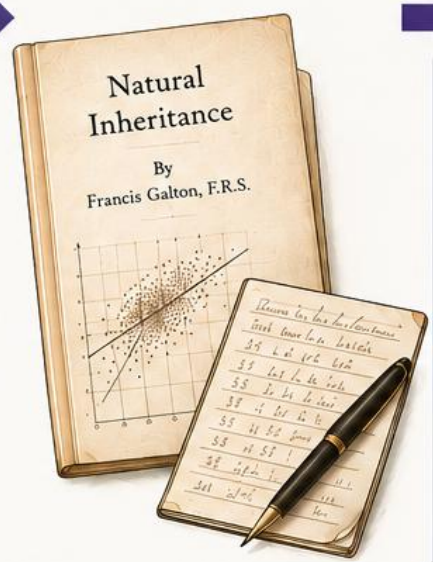


THE BIRTH OF A TERM

To describe this pattern, Galton coined the term

"REGRESSION TOWARD MEDIOCRITY"
(or toward the mean)

He described it mathematically and showed it with data, well before the era of modern statistics.

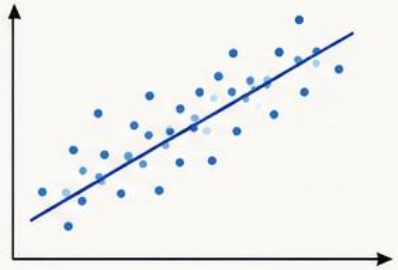


HOW THE IDEA EVOLVED

Galton's idea traveled far beyond biology.

In the 20th century, it became the foundation of regression analysis in statistics.

Today, regression is a fundamental tool for understanding relationships between variables in economics, medicine, social sciences, and many other fields.



$$Y = \beta_0 + \beta_1 X + \varepsilon$$

From studying pea seeds and dogs to building models that explain the world.



KEY INSIGHT

Extreme parents tend to have less extreme children – but the population's overall spread stays the same.

Definition of the simple regression model

Simple linear regression model

$$y = \beta_0 + \beta_1 x + u$$

Dependent variable (or explained variable, regressand) y.

Independent variable (or explanatory variable, regressor) x.

Error term (or disturbance) u captures unobserved factors affecting y ($E(u) = 0$).

Intercept parameter (or constant term) β_0.

Slope parameter β_1 …

… captures **ceteris paribus** effect Δx on y :$\Delta y = \beta_1 \Delta x$ if Δu = 0.

Obtaining the ordinary least squares (OLS) estimates

We would like to estimate β_0 and β_1 using a random sample $\{(x_i, y_i): i = 1, \dots, n\}$ from the population, with $y_i = \beta_0 + \beta_1 x_i + u_i$ for each i.

Here, u_i is the error term for observation i because it contains all factors affecting y_i other than x_i.

The estimated version of the population regression function is called the **sample regression function**.

$$\hat{y}_i = \hat{\beta}_0 + \hat{\beta}_1 x_i$$

Here, $\hat{\beta}_0$ and $\hat{\beta}_1$ are sample estimates of the population parameters β_0 and β_1, and $\hat{y}_i$ denotes the predicted value of y_i.

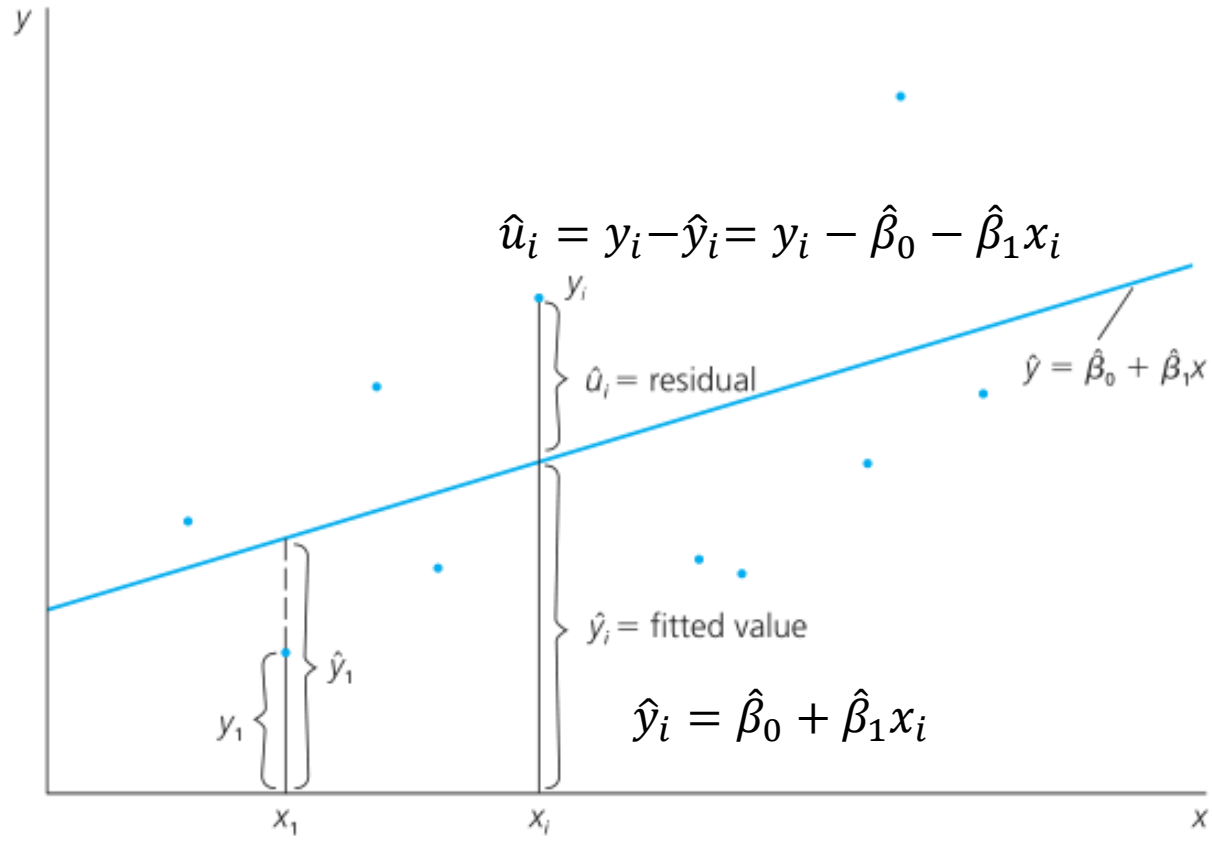
OLS regression line

The method of ordinary least squares (OLS) chooses the estimates to minimize the sum of squared residuals.

That is, given n observations on y_i and x_i the estimates $\hat{\beta}_0$ and $\hat{\beta}_1$ are chosen simultaneously to minimize:

$$\sum_{i=1}^{n} \hat{u}_i^{\,2} = \sum_{i=1}^{n} (y_i - \hat{\beta}_0 - \hat{\beta}_1 x_i)^2$$

Figure 1. Fitted values and residuals



Estimated parameters and their interpretation

The estimated slope parameter:

$$\hat{\beta}_1 = \frac{\sum_{i=1}^{n} (x_i - \bar{x})\,(y_i - \bar{y})}{\sum_{i=1}^{n} (x_i - \bar{x})^2}$$

The estimated intercept:

$$\hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x}$$

where $\bar{y}$ is the sample average of the y_i and likewise for $\bar{x}$.

The estimate of $\hat{\beta}_1$ has partial effect (or ceteris paribus) interpretations

The coefficient $\hat{\beta}_1$ measures the change in $\hat{y}$ due to a one-unit increase in x, holding all other factors fixed:

$$\Delta\hat{y} = \hat{\beta}_1 \Delta x$$

HOW GAUSS FOUND CERES

and invented the method of least squares

1 1 January 1801

On the first night of the nineteenth century, Italian astronomer Giuseppe Piazzi spotted something moving against the fixed stars.

Astronomers had been hunting for years for a "missing planet" between Mars and Jupiter.

He named it Ceres.



2 He tracked it for forty-one nights.

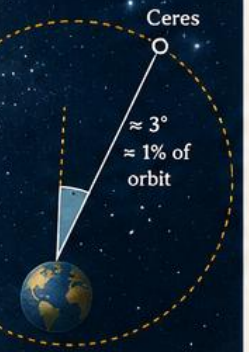
Then it drifted too close to the Sun, disappeared into the glare — and he fell ill.

When the sky cleared months later, nobody could find it again.

3 A tiny glimpse of a vast orbit

Piazzi had watched Ceres cross only about three degrees of sky, roughly one percent of its orbit.

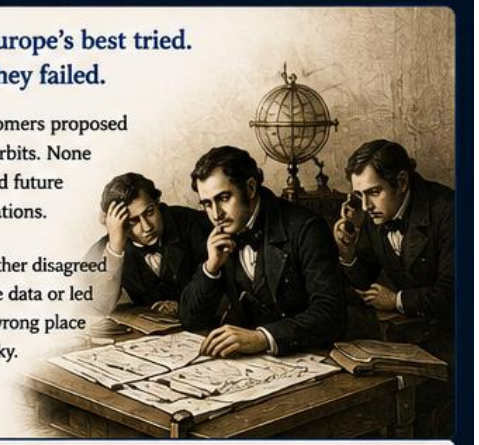
From that sliver, astronomers had to predict where an object would reappear across the whole celestial sphere, months later.



4 Europe's best tried. They failed.

Astronomers proposed many orbits. None matched future observations.

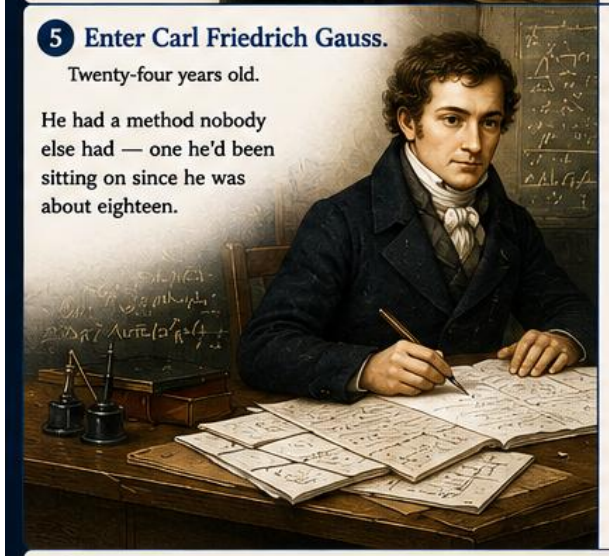
They either disagreed with the data or led to the wrong place in the sky.



5 Enter Carl Friedrich Gauss.

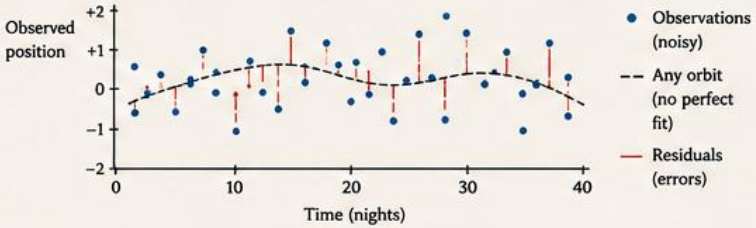
Twenty-four years old.

He had a method nobody else had — one he'd been sitting on since he was about eighteen.



6 Here's the problem he faced. And it's our problem.

Piazzi's observations were more numerous than the unknowns in the orbit, and they contradicted each other — each one carried small measurement errors. There was no orbit that fit all the data exactly.



Most astronomers:

Throw out observations, keep just enough to solve the equations cleanly.

Gauss:

Keep everything, accept that no curve will pass through every point, and ask a different question:

Which orbit makes the sum of squared errors as small as possible?

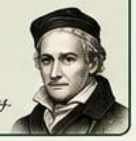
7 He computed for weeks. He sent a prediction.

On the last night of 1801, Franz Xaver von Zach pointed his telescope where Gauss said to look — and Ceres was there.



That made Gauss famous across Europe overnight.

But the method mattered more than the planet.



8 THE METHOD WE USE TODAY

You have noisy data, more observations than parameters, no perfect fit — so you find the line (or curve) that minimises the sum of squared residuals.



Noisy data → Best-fitting line (least squares) → Smallest sum of (squared residuals)

9 MATHEMATICALLY

Choose parameters β that minimise the sum of squared residuals (SSR):

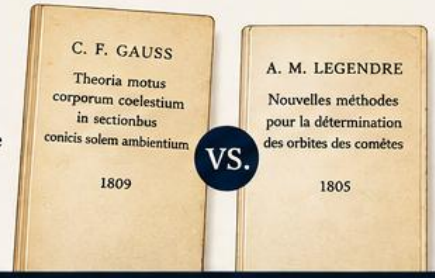
$$SSR = \sum_{i=1}^{n} (y_i - \hat{y}_i)^2$$

where y_i = observed value,
$\hat{y}_i$ = model prediction.

10 HISTORY OF THE IDEA

Gauss published the method in 1809.

But Legendre had published the same idea independently in 1805 — which sparked one of the great priority disputes in the history of mathematics.



THE BIG IDEA: When data are noisy and imperfect, the best we can do is find the explanation that makes the errors as small as possible.

Simple regression model example. Wage and education

$y = wage,$ the wage in dollars per hour of a person from a sample ($n = 526$).
$x = educ$, years of schooling of a person.

OLS regression line (or sample regression function)

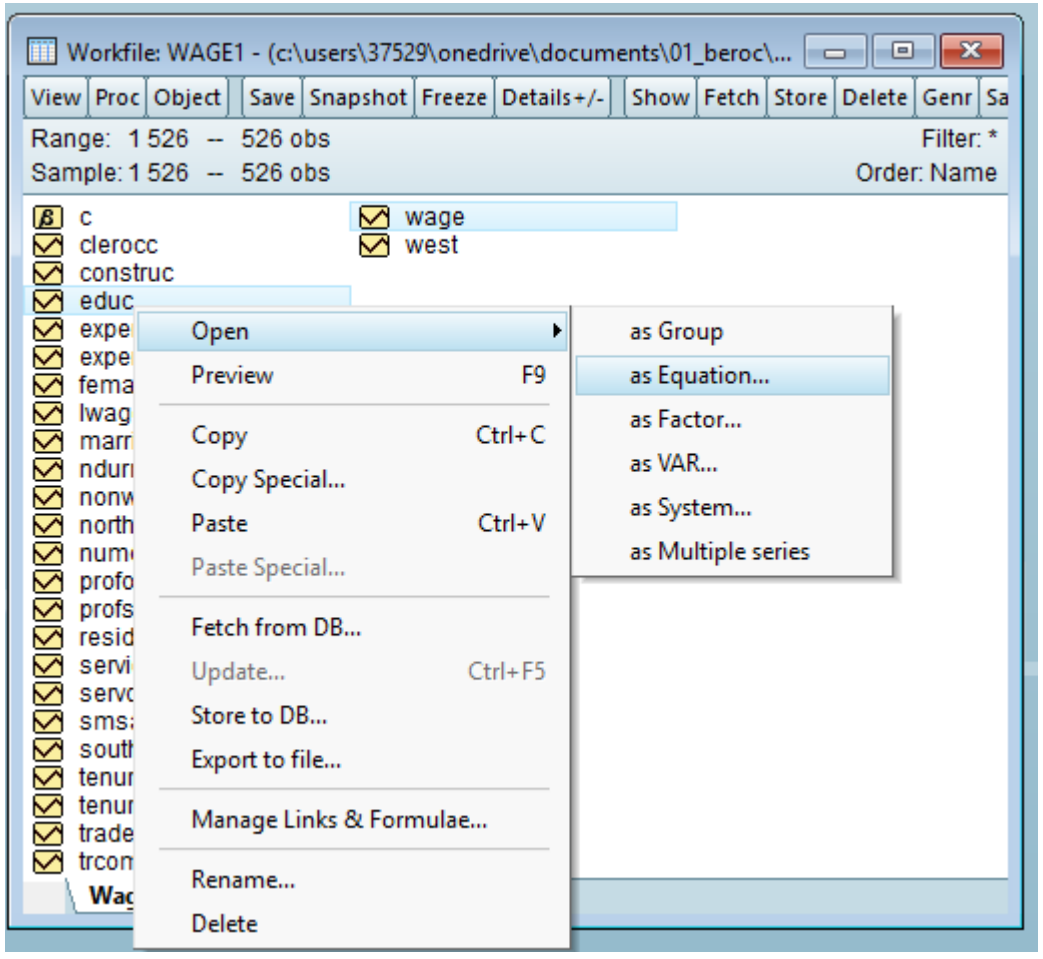
$$\widehat{wage} = -0.90 + 0.54\, educ$$

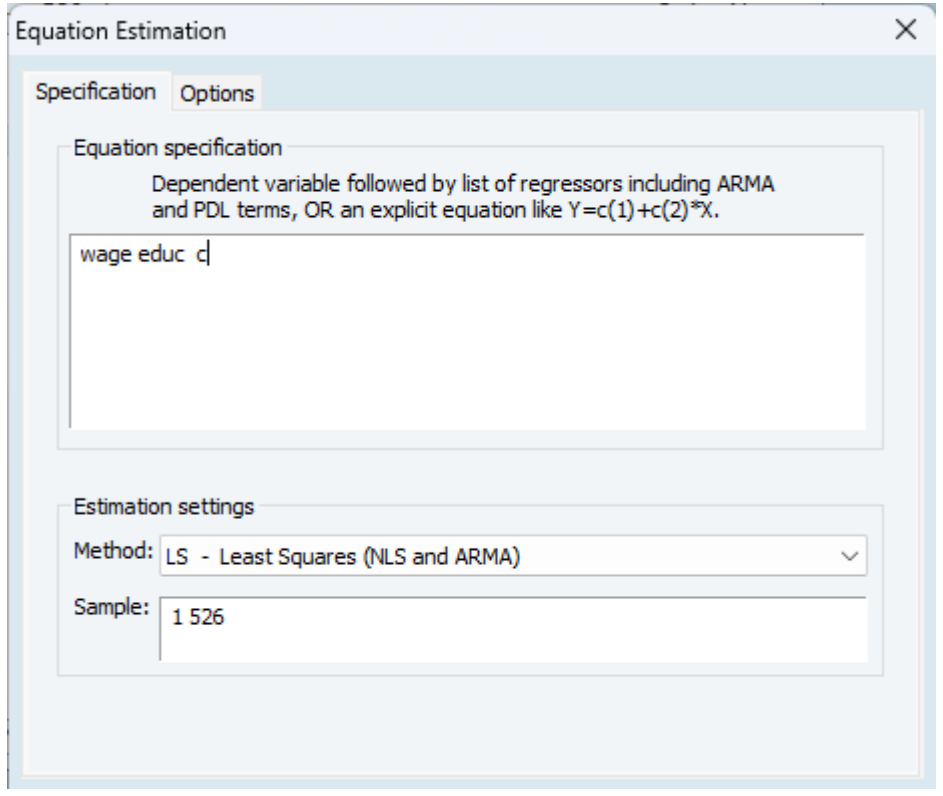
The intercept of –0.90 means that a person with no education has a predicted hourly wage of –90¢ an hour. This is silly.

The slope estimate implies that one more year of education increases hourly wage by 54¢ an hour.

Wage and education. EViews (1)

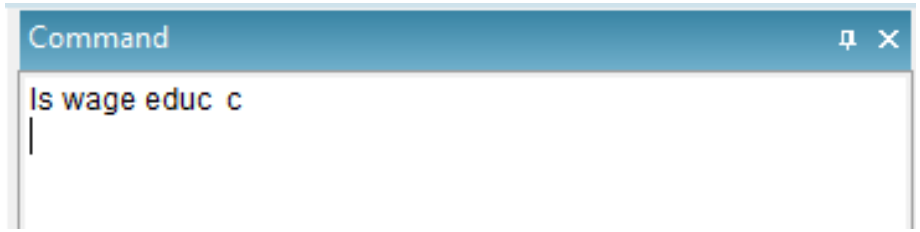
First select the dependent variable, then the explanatory variables (then – right click).





Wage and education. EViews (2)

... or use the command window



Dependent Variable: WAGE
Method: Least Squares
Date: 12/27/25 Time: 15:50
Sample: 1 526
Included observations: 526

Slope estimate
$\hat{\beta}_1$

Variable	Coefficient	Std. Error	t-Statistic	Prob.
EDUC	0.541359	0.053248	10.16675	0.0000
C	-0.904852	0.684968	-1.321013	0.1871

Intercept estimate
$\hat{\beta}_0$

R-squared	0.164758	Mean dependent var	5.896103
Adjusted R-squared	0.163164	S.D. dependent var	3.693086
S.E. of regression	3.378390	Akaike info criterion	5.276470
Sum squared resid	5980.682	Schwarz criterion	5.292688
Log likelihood	-1385.712	Hannan-Quinn criter.	5.282820
F-statistic	103.3627	Durbin-Watson stat	1.823686
Prob(F-statistic)	0.000000		

2. Multiple regression model

Motivation for multiple regression

Multiple regression analysis allows us to explicitly control for factors that simultaneously affect the dependent variable, as in

$$wage = \beta_0 + \beta_1 educ + \beta_2 exper + \beta_3 tenure + u$$

where $educ$ – years of education, $exper$ – years of labor market experience, and $tenure$ – years with the current employer.

The **general multiple linear regression model** (also called the *multiple regression model*) can be written in the population as

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_3 x_3 + \cdots + \beta_k x_k + u$$

where β_0 is the intercept; β_1 is the parameter associated with x_1; β_2 is the parameter associated with x_2, and so on.

Obtaining the OLS estimates

In the general case with *k* independent variables, we seek estimates $\hat{\beta}_0, \hat{\beta}_1, \ldots, \hat{\beta}_k$ in the equation (**OLS regression line** or the **sample regression function**).

$$\hat{y} = \hat{\beta}_0 + \hat{\beta}_1 x_1 + \hat{\beta}_2 x_2 + \cdots + \hat{\beta}_k x_k$$

The OLS estimates, *k+1* of them, are chosen to minimize the sum of squared residuals.

$$\sum_{i=1}^{n} \hat{u}_i^{\,2} = \sum_{i=1}^{n} (y_i - \hat{y}_i)^2 = \sum_{i=1}^{n} \left(y_i - \hat{\beta}_0 - \hat{\beta}_1 x_{i1} - \cdots - \hat{\beta}_k x_{ik}\right)^2$$

Here, $\hat{\beta}_0$ is the OLS intercept and $\hat{\beta}_1, \ldots, \hat{\beta}_k$ are the OLS slope estimates (corresponding to the explanatory variables $x_1, \ldots, x_k$).

Interpreting the OLS regression equation

The **OLS regression line** is

$$\hat{y} = \hat{\beta}_0 + \hat{\beta}_1 x_1 + \hat{\beta}_2 x_2 + \cdots + \hat{\beta}_k x_k$$

The intercept $\hat{\beta}_0$ is the predicted value of y when $x_1, \ldots, x_k = 0$.

The coefficient on x_1 measures the change in $\hat{y}$ due to a one-unit increase in x_1, holding all other independent variables fixed. That is,

$$\Delta\hat{y} = \hat{\beta}_1 \Delta\, x_1$$

holding x_2, x_3 , …, x_k fixed. Thus, we have *controlled for* the variables x_2, x_3 , …, x_k when estimating the effect of x_1 on y . The other coefficients have a similar interpretation.

Interpreting the OLS regression equation. Example in EViews

Using the 526 observations on workers, we include *educ* (years of education), *exper* (years of labor market experience), and *tenure* (years with the current employer) in an equation explaining *wage*.

$$\widehat{wage} = -2.87 + 0.60educ + 0.02exper + 0.17tenure$$

Dependent Variable: WAGE
Method: Least Squares
Date: 12/27/25 Time: 19:45
Sample: 1 526
Included observations: 526

Variable	Coefficient	Std. Error	t-Statistic	Prob.
EDUC	0.598965	0.051284	11.67948	0.0000
EXPER	0.022340	0.012057	1.852849	0.0645
TENURE	0.169269	0.021645	7.820361	0.0000
C	-2.872735	0.728964	-3.940844	0.0001

R-squared	0.306422	Mean dependent var	5.896103
Adjusted R-squared	0.302436	S.D. dependent var	3.693086
S.E. of regression	3.084476	Akaike info criterion	5.098216
Sum squared resid	4966.303	Schwarz criterion	5.130652
Log likelihood	-1336.831	Hannan-Quinn criter.	5.110916
F-statistic	76.87317	Durbin-Watson stat	1.791222
Prob(F-statistic)	0.000000		

OLS fitted values and residuals

Recall that the **fitted** or **predicted** value of *y* for observation *i* is

$$\hat{y}_i = \hat{\beta}_0 + \hat{\beta}_1 x_{i1} + \hat{\beta}_2 x_{i2} + \cdots + \hat{\beta}_k x_{ik}$$

and that the corresponding **residual** for observation *i* equals

$$\hat{u}_i = y_i - \hat{y}_i$$

The sample average of the residuals is *zero*.

Goodness-of-fit

Recall that *SST* = *SSE* + *SSR*, with

$SST = \sum_{i=1}^{n}(y_i - \bar{y})^2$ the **total sum of squares**.

$SSE = \sum_{i=1}^{n}(\hat{y}_i - \bar{y})^2$ the **explained sum of squares**.

$SSR = \sum_{i=1}^{n} \hat{u}_i{}^2$ the **residual sum of squares** (or sum of sq. res.).

The **R-squared** (R^2 or coefficient of determination)

$$R^2 = \frac{SSE}{SST} = 1 - \frac{SSR}{SST}$$

$0 \leq R^2 \leq 1$, when an intercept is included *(can be negative in the model without intercept)*.

measures the fraction of the sample variation in y explained by $\hat{y}$.

R-squared

$$\widehat{wage} = -2.87 + 0.60educ + 0.02exper + 0.17tenure$$

R^2 never decreases (and usually increases) when another independent variable is added to a regression.

Thus, one should be careful when choosing the best model based on R^2 only.

Regression with low R^2 is common in the social sciences, but may estimate ceteris paribus relation well.

Dependent Variable: WAGE
Method: Least Squares
Date: 12/27/25 Time: 19:45
Sample: 1 526
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F-statistic	76.87317	Durbin-Watson stat	1.791222
Prob(F-statistic)	0.000000		

Adjusted R-squared is adjusted for the number of regressors (or adjusted for the degrees of freedom).

3. Assumptions of the classical linear regression model

Assumption MLR.1. Linear in parameters

The model in the population can be written as

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \cdots + \beta_k x_k + u$$

where $\beta_0, \beta_1, \beta_2, \ldots, \beta_k$ are the unknown parameters of interest and u is an unobserved random error or disturbance term.

Still, OLS is quite flexible because one can use arbitrary functions of the underlying variables of interest (natural logarithms, squares, etc.).

Thus, OLS cannot estimate coefficient $\beta^3 x$ but can estimate βx^3 or $\beta \ln(x)$, etc.

Assumption MLR.2. Random sampling

We have a random sample of *n* observations, $\{(x_{i1}, x_{i2}, \dots, x_{ik}, y_i): i = 1, 2, \dots, n\}$, following the population model in Assumption MLR.1.

Random sampling is what licenses generalising from the sample to the population.

Assumption MLR.3.
No perfect collinearity

In the sample (and therefore in the population), none of the independent variables is constant, and there are **no exact linear relationships** among the independent variables.

The Assumption MLR.3 *does* allow the independent variables to be correlated; they just cannot be *perfectly* correlated.

This model violates Assumption MLR.3 because $x_3 = x_1 + x_2$ by definition.

$$voteA = \beta_0 + \beta_1 expendA + \beta_2 expendB + \beta_3 totexpend + u,$$

where *voteA* is the percentage of the vote for Candidate A, *expendA* are campaign expenditures by Candidate A, *expendB* are campaign expenditures by Candidate B, and totexpend are total campaign expenditures.

Assumption MLR.4.
Zero conditional mean

The error u has an expected value of zero given any values of the independent variables.

$$\mathrm{E}(u|x_1, x_2, \dots, x_k) = 0$$

One way that Assumption MLR.4 can fail is if the functional relationship between the explained and explanatory variables is misspecified. If the true model has *log(wage)* as the dependent variable but we use *wage*, then the estimators will be biased.

Omitting an important factor that is correlated with any of the independent variables causes Assumption MLR.4 to fail also.

Unfortunately, this assumption cannot be tested!

Unbiasedness of OLS

Under assumptions MLR.1 – MLR.4 the OLS estimators are **unbiased estimators** of the population parameters

$$E(\hat{\beta}_j) = \beta_j, \qquad j = 0, 1, \dots, k.$$

This does not mean, however, that estimates from a single sample are unbiased but rather that the procedure produces unbiased results.

Omitted variable bias

Population model $wage = \beta_0 + \beta_1 educ + \beta_2 ability + u$

Simple regression $\widetilde{wage} = \tilde{\beta}_0 + \tilde{\beta}_1 educ$

$$\tilde{\beta}_1 = \hat{\beta}_1 + \hat{\beta}_2 \tilde{\delta}_1$$

where $\hat{\beta}_1$ and $\hat{\beta}_2$ are the slope estimators (if we could have them) from the multiple regression of y_i on $x_{i1}, x_{i2}, \quad i = 1, \dots, n$

and $\tilde{\delta}_1$ is the slope from the simple regression of x_{i2} on $x_{i1}, \quad i = 1, \dots, n$

The bias (**omitted variable bias**) in $\tilde{\beta}_1$ is $\text{Bias}(\tilde{\beta}_1) = E(\tilde{\beta}_1) - \beta_1 = \beta_2 \tilde{\delta}_1$

Assumption MLR.5. Homoskedasticity

Question: What forecast is better (i) 2 to 4% GDP growth or (ii) 1 to 5% GDP growth?

Homoskedasticity

The error u has the same variance given any values of the explanatory variables.

$$\mathrm{Var}(u|x_1, x_2, \ldots, x_k) = \sigma^2$$

Sampling variances of the OLS slope estimators

Under Assumptions MLR.1 through MLR.5, conditional on the sample values of the independent variables,

$$\mathrm{Var}(\hat{\beta}_j) = \frac{\sigma^2}{\mathrm{SST}_j(1 - \mathrm{R}_j^2)},$$

for $j = 1, 2, \ldots, k$, where $SST_j = \sum_{i=1}^{n}(x_{ij} - \bar{x}_j)^2$ is the total sample variation in x_j, and R_j^2 is the R-squared from regressing x_j on all other independent variables (and including an intercept).

Gauss-Markov theorem

Assumptions MLR.1 – MLR.5 are referred to as the Gauss-Markov assumptions (for cross-sectional regression).

There may be many unbiased estimators satisfying assumptions MLR.1 through MLR.4.

However, if we also add assumption MLR.5 then OLS is the **best linear unbiased estimator** (BLUE) – Gauss-Markov theorem

Best is defined as *having the smallest variance*. Given two unbiased estimators, it is logical to prefer the one with the smallest variance.

Regression through the origin

OLS regression through the origin

$$\tilde{y} = \tilde{\beta}_1 x_1 + \tilde{\beta}_2 x_2 + \cdots + \tilde{\beta}_k x_k$$

where the symbol "~" over the estimates is used to distinguish them from the OLS estimates obtained along with the intercept.

The OLS residuals no longer have a zero sample average.

R^2 can actually be negative. The sample average, $\bar{y}$, "explains" more of the variation in the y_i than the explanatory variables.

If the intercept β_0 in the population model is different from zero, then the OLS estimators of the slope parameters (through the origin) will be biased.

The cost of estimating an intercept when β_0 is truly zero is that the variances of the OLS slope estimators are larger.

4. Units of measurement and functional form

Changing units of measurement

Salary in $ thousands

ROE in %

Estimation Command:
==========
LS SALARY ROE C

Estimation Equation:
==========
SALARY = C(1)*ROE + C(2)

Substituted Coefficients:
==========
SALARY = 18.50*ROE + 963.19

Salary in $

ROE in %

Estimation Command:
==========
LS SALARY*1000 ROE C

Estimation Equation:
==========
SALARY*1000 = C(1)*ROE + C(2)

Substituted Coefficients:
==========
SALARY*1000 = 18501.2*ROE + 963191.3

Salary in $ thousands

ROE in % / 100

Estimation Command:
==========
LS SALARY ROE/100 C

Estimation Equation:
==========
SALARY = C(1)*ROE/100 + C(2)

Substituted Coefficients:
==========
SALARY = 1850.12*ROE/100 + 963.19

Incorporating nonlinearities. Logarithm of the dependent variable

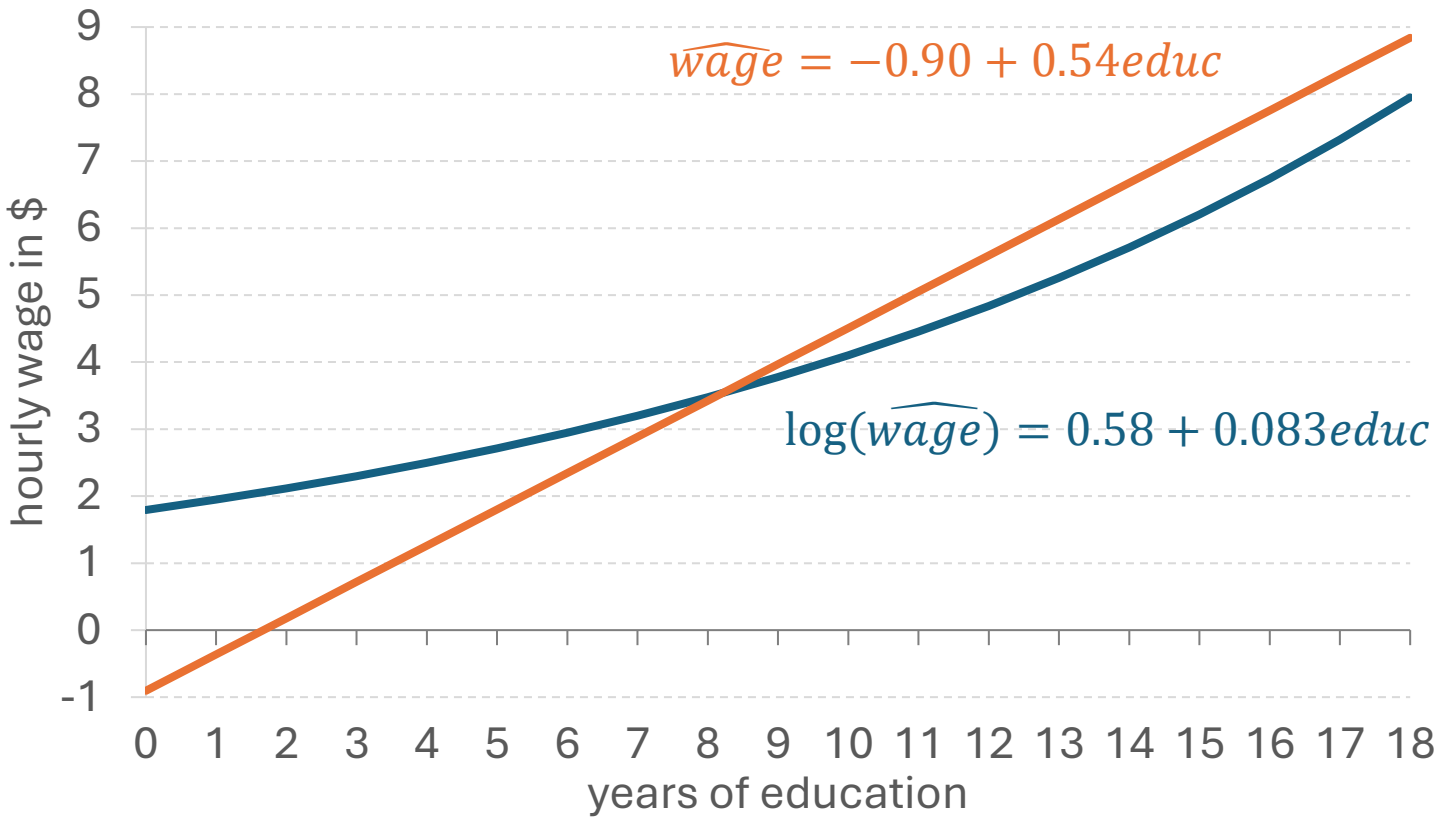
Wage and education

Substituted Coefficients:
==========
WAGE = 0.54*EDUC - 0.90

Natural logarithm of wage and education

Substituted Coefficients:
==========
LOG(WAGE) = 0.083*EDUC + 0.58

$$\%\Delta wage \approx (100 * \beta_1)\Delta educ$$



Wage increases by 8.3% for every additional year of education.

Incorporating nonlinearities. Constant elasticity model

Salary and sales

Substituted Coefficients:
==========
SALARY = 0.015*SALES + 1174.0

Natural logarithm of both salary and sales

Substituted Coefficients:
==========
LOG(SALARY) = 0.257*LOG(SALES) + 4.82

A 1% increase in firm sales increases CEO salary by about 0.257%.

Summary of functional forms involving logarithms

Model	Dependent Variable	Independent Variable	Interpretation of β_1
Level-level	y	x	$\Delta y = \beta_1 \Delta x$
Level-log	y	$\log(x)$	$\Delta y = (\beta_1/100)\%\Delta x$
Log-level	$\log(y)$	x	$\%\Delta y = (100\beta_1)\Delta x$
Log-log	$\log(y)$	$\log(x)$	$\%\Delta y = \beta_1 \%\Delta x$

Semi-elasticity

Elasticity

Summary (1)

1. The multiple regression model allows us to effectively hold other factors fixed while examining the effects of a particular independent variable on the dependent variable. It explicitly allows the independent variables to be correlated.

2. Although the model is linear in its parameters, it can be used to model nonlinear relationships by appropriately choosing the dependent and independent variables.

3. The method of ordinary least squares is easily applied to estimate the multiple regression model. Each slope estimate measures the partial effect of the corresponding independent variable on the dependent variable, holding all other independent variables fixed.

4. R^2 is the proportion of the sample variation in the dependent variable explained by the independent variables, and it serves as a goodness-of-fit measure. It is important not to put too much weight on the value of R^2 when evaluating econometric models.

Summary (2)

5. Under the first four Gauss-Markov assumptions (MLR.1 through MLR.4), the OLS estimators are unbiased. This implies that including an irrelevant variable in a model has no effect on the unbiasedness of the intercept and other slope estimators. On the other hand, omitting a relevant variable causes OLS to be biased. In many circumstances, the direction of the bias can be determined.

6. Adding an irrelevant variable to an equation generally increases the variances of the remaining OLS estimators because of multicollinearity.

7. Under the five Gauss-Markov assumptions (MLR.1 through MLR.5), the OLS estimators are the best linear unbiased estimators (BLUEs).